

Tanzim Ikram Sheikh

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[Google Scholar](#) | [ORCID](#) | [Website](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

American International University–Bangladesh (AIUB)

Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering (CSE)

June 2021 – December 2025

Thesis: "Integrating Therapy and Technology: An Exploratory Case Study on the Development of Aids for Dementia"

CGPA: 3.82/4.00

RESEARCH INTERESTS

Human-Computer Interaction, Tangible Interaction, Multimodal and Gesture-Based Interfaces, Haptic Feedback Systems, Assistive Technologies for Aging and Dementia, Smart Healthcare Systems, Human-Robot Interaction, Extended Reality (XR), Human-Centered AI, Human-Centered Design, User Experience Design, Embedded Systems.

PUBLICATIONS

- **T. I. Sheikh**, M. R. Al Sultan, F. Ahmed, M. M. Iqra, and K. T. Hasan. Integrating Therapy and Technology: An Exploratory Case Study on the Development of Aids for Dementia in Bangladesh. ITAP 2026 (HCII), Lecture Notes in Computer Science (LNCS) 2026, 15903.
- S. M. Rohan, **T. I. Sheikh**, F. B. Zaman, M. Ahmed, M. S. Kabir, and K. T. Hasan. ReedMap: A Tangible Interaction System for Context-Aware and Division-Based Map Navigation. DUXU 2026 (HCII), Lecture Notes in Computer Science (LNCS) 2026, 15893.
- **T. I. Sheikh**, S. M. Rohan, N. Aziz, A. Amin, S. F. Mahbub, and K. T. Hasan. AirTune: A Tangible Interaction System for Gesture-Based and Contact-less Media Control. HCI 2026 (HCII), Lecture Notes in Computer Science (LNCS) 2026, 15899.
- K. Redwan, M. Ahmed, **T. I. Sheikh**, K. M. S. Hasan, M. F. A. Al Sohan, and K. T. Hasan. GlobeStick: Innovating Map Navigation Through Tactile Interaction and User-Centered Design. DUXU 2025 (HCII), Lecture Notes in Computer Science (LNCS) 2025, 15798.
- M. Ahmed, A. Sultana, P. B. Pew, S. Datto, **T. I. Sheikh**, K. Redwan, and M. F. A. Al Sohan. Melanoma Detection Using Augmented ResNet34 for High-Precision Dermoscopic Image Classification. ICCIES 2025, Lecture Notes in Networks and Systems (LNNS) 2025.

WORK EXPERIENCE

The Abroad Company

Remote

Product Experience Engineer

December 2025 – September 2026

• AbroadMates

- Designed the mentor landing page and streamlined the mentor signup flow, achieving a 100% mentor profile completion rate.
- Built a comprehensive mentor onboarding system, reducing drop-off and improving activation across the platform.
- Redesigned student and mentor dashboards with responsive data visualization and clearer information architecture, improving cross-device usability.
- Architected end-to-end user flows for an algorithmic mentor recommendation engine and a concurrent multi-mentor booking system with automated bundle discounts (e.g., 10% off for 5 mentors).
- Improved mentor discoverability by redesigning profile cards and detailed mentor pages.

• ApplicationMate

- Contributed to the foundation of a student-focused digital platform aimed at simplifying higher-education decision making through thoughtful interface design and workflow planning.
- Established the product's design direction by building a unified design system spanning onboarding, dashboards, and core workflows.
- Designed decision-support interfaces for country, subject, and program recommendations, enabling users to navigate complex study-abroad choices through clearer, more structured guidance.
- Designed the centralized student profile so applicants can enter data once and reuse it across applications.

Envolead Ltd.

Hybrid

Digital Innovation & Impact Intern

May 2025 – October 2025

- Built and deployed a full-stack company website, including admin dashboard, and CMS, using Laravel, Vue.js, and MySQL.
- Designed EnvoYouth, a mobile platform connecting youth climate activists, researchers, and sustainability advocates, and established a consistent design system aligned with the platform's environmental mission and brand guidelines.
- Automated personalized invitation generation and bulk email distribution using Python, enabling outreach at scale while saving significant time and reducing repetitive manual effort for the EnvoYouth launching ceremony.

- Developed a real-time graphical interface for telemetry monitoring, data logging, and diagnostics of CanSat and flight computer systems.
- Built an ESP8266-based web server interface for the AERD Launch Box to support hybrid rocket fuel grain testing and monitoring.
- Collaborated across multidisciplinary engineering teams spanning aerospace experimentation, embedded systems, and interface development.

PROJECTS

- **GyroWheel:** A BLE steering controller that uses an ESP32 and MPU6050 IMU to enable tilt-based steering for racing games, with tactile buttons mapped to essential driving controls such as throttle, brake, nitro boost, and camera switching.
- **ShooterX:** An ESP32-powered customizable FPS gaming controller integrating MPU6050 motion tracking, rotary encoders, touch sensors, and Bluetooth HID communication. Designed to enhance immersive gameplay through gyroscopic aiming, haptic feedback, and modular physical controls.

TECHNICAL SKILLS

- **Design:** Figma, User Research, Usability Testing, Wireframing, Prototyping, Design Systems, User Flows, Information Architecture
- **Programming Languages:** Python, JavaScript/TypeScript, PHP, C/C++
- **Research & Data Tools:** NumPy, Pandas, Matplotlib, R
- **Web Frameworks:** Next.js, React, Node.js, Nest.js
- **Embedded Systems:** Arduino, ESP32/ESP8266, Sensor Integration, Circuit Design
- **Tools:** Git, Postman, Adobe Creative Suite, EasyEDA, Overleaf, VS Code, Jupyter Notebook, Google Colab

UX RESEARCH CASE STUDIES

Namjari App (Government Land Mutation App)

Team Lead, Group UX Case Study

Led a cross-functional team to redesign Bangladesh’s land mutation process into a guided, one-stop digital solution with unified payments and an AI chatbot, "Bhumisathi." Directed usability testing that reduced task completion time by 60% and raised task success rate from baseline to 100%, applying accessibility and trust-centered design principles for citizen users.

NID Wallet App (Government Digital ID App)

Team Lead, Group UX Case Study

Directed the redesign of the NID Wallet app into NID Sheba, a unified digital ID solution built around competitive analysis and a trust-centric onboarding flow with multilingual support. Led usability testing that improved task success from 60% to 100% and cut completion time by approximately 80%, and integrated biometric verification to strengthen security and user trust.

HONOURS AND AWARDS

1st Place (Joint Winner), Group UX Case Study Competition Ostad Limited	2025
1st Runner-Up, Logo Design Competition, AIUB CS Fest AIUB	2024
Leadership Award, English Olympiad Season 2 English Olympiad	2019

LANGUAGE PROFICIENCY

Academic IELTS	Overall: 7.5 CEFR Level: C1
Listening: 8.5, Reading: 8.5, Writing: 6.5, Speaking: 6.5	

EXTRA-CURRICULAR & VOLUNTEERING ACTIVITIES

- Participant, Line Following Robot Competitions
 - Tech Envoy, Tech Topia
 - Member, Roboment
- General Member, AIUB Environment Club
 - Senior Member, Connecting Youth for Change
 - Campus Ambassador, English Olympiad

REFERENCES

Dr. Khandaker Tabin Hasan Professor & Associate Dean Department of Computer Science Faculty of Science and Technology American International University–Bangladesh tabin@aiub.edu	Md. Mehzabul Hoque Nahid Senior Assistant Professor & Department Head Department of Management Information System Faculty of Business Administration American International University–Bangladesh nahid@aiub.edu
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